

I have integrated three major principles of quantum physics into the standard 8x8 chess framework:

- **Superposition (The Split Move):** Using the '**S**' key, players can split a piece into two "Ghost" states. The piece technically exists in both squares simultaneously until it is "measured."
- **Entanglement:** If a player moves a piece into a square occupied by another piece's "Ghost," the two pieces become red-linked. A change or collapse in one instantly forces a collapse in the other, regardless of distance.
- **Quantum Tunneling:** Sliding pieces (Rooks, Bishops, Queens) have a **30% probability** to pass through solid obstacles. This models how subatomic particles can "leak" through barriers that are classically impassable.

Technical Features & UI

- **Dynamic Educational Sidebar: * The Handbook:** A permanent guide explaining the physics behind the moves.
 - **Live "Did You Know?":** A fact-ticker that changes **after every move**, providing real-world examples of quantum tech (MRIs, Lasers, etc.).
 - **Live Inventory:** A real-time counter showing exactly how many White and Black pieces remain.
- **The "Collapse" Engine:** An automated logic system that handles **Measurement**. When a player lands on a Ghost square, the game "observes" the state and randomly collapses the superposition into a single reality.
- **Win State Logic:** A dedicated **Win Screen** that triggers when a King is captured, stopping all game play and offering a "Restart" option.

Visuals & Stability

- **Transparency Handling:** I have implemented `.convert_alpha()` logic to ensure PNG chess pieces look professional without ugly background boxes.
- **Knight (Horse) Logic:** I have custom-coded the Knight's movement to ensure it maintains its "L-shape" jump while ignoring the Tunneling probability (since Knights naturally jump).
- **Entanglement Rings:** Visual red indicators that show players which pieces are currently "spookily" linked.
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AI Integration & Development Role

I utilized **Gemini** as a collaborative development partner to solve the unique challenges of merging quantum mechanics with classical game logic. The AI's role included:

- **Logic Synthesis:** The AI helped translate abstract physics concepts into functional Python algorithms. For example, it assisted in creating the **Entanglement Logic**, ensuring that a `collapse()` function called on one object would trigger a recursive check for its entangled partner.
- **Probabilistic Engine Design:** I worked with the AI to implement the **Tunneling Mechanism**.
- **Debugging & Asset Management:** The AI provided the technical solution for **Transparency Handling** (using `.convert_alpha()`), which fixed issues with non-transparent PNG backgrounds, and assisted in debugging the **Knight's L-shape jumping logic** to ensure it bypassed the tunneling probability check.